

PRESENTATION SLIDE STRUCTURE

AI Drowsiness Alert System

Real-Time Driver Fatigue Detection Using Computer Vision

SLIDE 1 — TITLE PAGE

AI Drowsiness Alert System

Real-Time Driver Fatigue Detection Using Computer Vision

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SLIDE 2 — INTRODUCTION

 **Introduction**

Driver fatigue and micro-sleep are major causes of road accidents worldwide.

Many accidents occur because drivers become sleepy without realizing it.

This project was developed to create a real-time intelligent monitoring system that detects signs of drowsiness using:

Artificial Intelligence (AI)

Computer Vision

Facial Landmark Detection

The system monitors eye behavior through a webcam and alerts the driver before falling asleep.

SLIDE 3 — SITUATION (STAR METHOD)

 **Situation**

Many road accidents are caused by:

Driver fatigue

Long driving hours

Lack of sleep

Micro-sleep during driving

There was a need for a system that could:

Monitor the driver in real time

Detect signs of drowsiness early

Warn the driver before dangerous situations occur

SLIDE 4 — TASK (STAR METHOD)

Task

The main objectives of this project were to:

Access live webcam video

Detect facial landmarks in real time

Isolate eye regions

Calculate Eye Aspect Ratio (EAR)

Detect eye closure duration

Differentiate blinking from drowsiness

Trigger an alarm during prolonged eye closure

Additional Features Implemented

☒ **Adaptive Thresholding**

☒ **Timer-Based Detection**

☒ **Real-Time Audio Alert System**

SLIDE 5 — TECHNOLOGIES & WORKFLOW

Technologies Used

Programming Language

Python

Libraries Used

OpenCV

MediaPipe

NumPy

SciPy

Playsound

Concepts Applied

Computer Vision

Facial Landmark Detection

Feature Extraction

Real-Time Video Processing

Eye Aspect Ratio (EAR)



System Workflow

Webcam Input---Face Detection---Facial Landmark Mapping---Eye Landmark Extraction--EAR Calculation---Adaptive Threshold Calibration---Eye Closure Detection ---Timer-Based Validation---Alarm Trigger

SLIDE 6 — ACTION (PART 1)



Action — Facial Landmark Detection

The system uses:

OpenCV for webcam access

MediaPipe Face Mesh for facial landmark detection

What the System Detects

468 facial landmarks

Real-time eye landmark coordinates

Facial movement tracking

The webcam continuously monitors the driver's face and eye movement.

 (Insert Facial Landmark Screenshot Here)

SLIDE 7 — ACTION (PART 2)

Action — EAR Calculation

The Eye Aspect Ratio (EAR) was used to determine whether the eyes are open or closed.

EAR Formula

EAR Interpretation

Eye State EAR Behavior
☺ Eyes Open Higher EAR values
👁️ Blinking Temporary EAR drop
☹️ Eyes Closed EAR drops below threshold

The EAR value changes dynamically depending on eye movement.

 (Insert EAR Screenshot Here)

SLIDE 8 — ACTION (PART 3)

Action — Adaptive Threshold & Timer Logic

To reduce false alarms and improve accuracy, we implemented:

- ☒ Adaptive Thresholding
- ☒ Timer-Based Eye Closure Detection

How It Works

Adaptive Thresholding

The system first observes the user's normal eye state

Multiple EAR samples are collected

A personalized threshold is generated automatically

Timer Logic

Short eye closure → Blink

Long eye closure (4+ seconds) → Drowsiness

This helped reduce false positives caused by normal blinking.

SLIDE 9 — ADAPTIVE THRESHOLD DEMONSTRATION

Adaptive Threshold Demonstration

The system first performs a calibration stage before monitoring begins.

During calibration:

Multiple EAR samples are collected

The average EAR is calculated

A personalized adaptive threshold is generated

This allows the system to adapt to:

Different eye shapes

Small eyes

Lazy eyes

Individual facial differences

Calibration Stages

Screenshot 1

CALIBRATING...KEEP EYES OPENCollecting Samples: 62/100Current EAR: 0.355

Screenshot 2

Calibration CompleteAverage EAR: 0.356Adaptive Threshold: 0.268

Screenshot 3

EAR: 0.141Threshold: 0.268

SLIDE 10 — RESULT

Result

Project Outcome

- ☒ **Successfully detected facial landmarks in real time**
- ☒ **EAR updated dynamically during monitoring**
- ☒ **Differentiated blinking from drowsiness**

- ☑ Triggered audio alarm during prolonged eye closure
- ☑ Worked successfully on standard laptop hardware

Final Output

The system can monitor a driver in real time and warn them before falling asleep.

🖼️ (Insert Drowsiness Alert Screenshot Here)

SLIDE 11 — REAL-WORLD TESTING & LIMITATIONS

▨ Real-World Testing

The project was tested under different real-world conditions to evaluate performance and reliability.

Successful Cases

- ☑ Bright environment
- ☑ Normal indoor lighting
- ☑ Direct face visibility

Challenging Cases

- ⚠️ Dark environments reduced accuracy
- ⚠️ Glasses sometimes affected eye landmark detection
- ⚠️ Partial face obstruction affected tracking

These observations helped identify areas for future improvement.

🖼️ (Insert Bright Environment Screenshot Here)

🖼️ (Insert Dark Environment Screenshot Here)

🖼️ (Insert Glasses Screenshot Here)

SLIDE 12 — LIMITATIONS & FUTURE IMPROVEMENTS

⚠️ Current Limitations

The current system mainly focuses on:

Eye behavior monitoring

EAR-based drowsiness detection

Existing Limitations

No yawning detection yet

No head pose estimation

Performance reduces under poor lighting

Webcam quality affects detection accuracy

Glasses or occlusion may reduce reliability

Future Improvements

Future versions may include:

Yawning detection

Head pose estimation

Deep learning-based fatigue classification

Mobile deployment

Edge AI deployment

Smart vehicle dashboard integration

Cloud-based monitoring systems

SLIDE 13 — DEPLOYMENT & CONCLUSION

 **Deployment & Real-World Applications**

This project belongs to the fields of:

Artificial Intelligence (AI)

Computer Vision

Human Monitoring Systems

Embedded Safety Systems

Possible Deployment Areas

Smart vehicles

Driver assistance systems

Transportation safety systems

Industrial machine monitoring

Edge AI cameras

Smart surveillance systems

Project Type

Real-Time AI & Computer Vision Monitoring System

The project combines:

AI-based facial analysis

Real-time video processing

Human behavior monitoring

Intelligent alert systems

Conclusion

This project demonstrates how Artificial Intelligence and Computer Vision can improve safety by detecting driver drowsiness in real time.

The system provides a strong foundation for future intelligent driver-assistance technologies.